Literature status of the Born–Jordan smoothing problem in arXiv:1507.00144

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Status. literature_already_answered. A separate paper by the same authors gives a quantitative microlocal smoothing theorem and a complementary global obstruction.

Original problem

For a signal f, its Born–Jordan distribution is

$$Qf = Wf * \Theta_\sigma, \qquad \Theta(\zeta_1, \zeta_2) = \frac{\sin(\pi\zeta_1 \cdot \zeta_2)}{\pi\zeta_1 \cdot \zeta_2}.$$

On source PDF page 2, Cordero–de Gosson–Nicola observe that multiplication by Θ visibly damps ghost frequencies but state that analytically quantifying this smoothing effect remains a challenging open problem [1]. The intended comparison is between the Wigner distribution Wf and the Born–Jordan distribution Qf.

The separate answer

The later paper [2] explicitly presents itself as a rigorous quantitative explanation of this phenomenon for arbitrary tempered signals. Let $WF_{\mathcal{F}L^q_s}$ denote the Fourier–Lebesgue wave-front set. Its Theorem 1.2 states that, if

$$f \in \mathcal{S}'(\mathbb{R}^d), \qquad Wf \in M^{\infty,q}(\mathbb{R}^{2d}),$$

then for every (z, ζ) with $\zeta = (\zeta_1, \zeta_2)$ and $\zeta_1 \cdot \zeta_2 \neq 0$,

$$(z, \zeta) \notin WF_{\mathcal{F}L^q_2}(Qf).$$

Thus Qf has two additional microlocal derivatives in every non-characteristic direction. Corollary 1.3 gives the particularly concrete case

$$f \in L^2(\mathbb{R}^d) \quad \Longrightarrow \quad (z, \zeta) \notin WF_{H^2}(Qf) \quad (\zeta_1 \cdot \zeta_2 \neq 0).$$

These statements occur on supporting PDF page 6.

The result is also sharp in the essential geometric sense. Theorem 1.4 proves that an estimate

$$\|Qf\|_{M^{p,q_1}} \le C\|Wf\|_{M^{p,q_2}} \qquad (f \in \mathcal{S})$$

forces $q_1 \ge q_2$. Hence there is no uniform global improvement of the modulation-space smoothness exponent. The exceptional set $\zeta_1 \cdot \zeta_2 = 0$ contains the horizontal and vertical directions that retain interference because the Born–Jordan transform preserves marginals.

Conclusion

The 2016/2018 paper answers the source's open problem in the natural sharp form: Born–Jordan provides a two-derivative microlocal gain away from the characteristic axes, while no general global smoothness gain can cross those axes. This is an explicit later-literature answer, not a new agent result.

References

[1] E. Cordero, M. de Gosson, and F. Nicola, *On the Invertibility of Born–Jordan Quantization*, J. Math. Pures Appl. 105 (2016), 537–557; arXiv:1507.00144.

[2] E. Cordero, M. de Gosson, and F. Nicola, *On the reduction of the interferences in the Born–Jordan distribution*, Appl. Comput. Harmon. Anal. 44 (2018), 230–245; arXiv:1601.03719; doi:10.1016/j.acha.2016.04.007.